A Comparative Study of CNN and Transformer Architectures for Plant Disease Classification

Paolo Bigliardi Niccolò Davoli

Abstract

This paper presents a comparative evaluation of six deep learning architectures on the task of plant disease classification using the PlantVillage dataset. We evaluate three foundational convolutional neural networks—AlexNet (2012), VGG-16 (2014), and ResNet-50 (2015)—trained from scratch, alongside three transformer-based models: DINOv2 (ViT-B/14) evaluated via linear probing on frozen self-supervised features, a Vision Transformer (ViT-B/16) fine-tuned from supervised ImageNet pre-training, and a CLIP-architecture ViT-B/16 trained from random initialisation. This design spans a broad spectrum of training paradigms—from end-to-end learning without pre-training, through self-supervised and supervised transfer learning, to training a modern architecture from scratch—enabling a controlled analysis of how architectural design and pre-training strategy affect classification performance. We evaluate each model across multiple metrics including top-1 and top-5 accuracy, balanced accuracy, macro-averaged precision, recall, and F1 score, per-class analysis, predictive entropy, model size, and computational cost. VGG-16 achieves the highest top-1 accuracy (99.38%), followed by ResNet-50 (98.86%), AlexNet (98.66%), DINOv2 via linear probing (98.37%), the fine-tuned ViT (98.00%), and the randomly initialised CLIP-architecture model (95.07%). The narrow performance gap among the top four models reflects the limited visual complexity of PlantVillage under controlled conditions, while the comparison between fine-tuned ViT and randomly initialised ViT-B/16 confirms that pre-training provides a consistent and measurable advantage. We discuss dataset limitations—particularly data leakage from correlated samples under random splitting—and the implications for interpreting the reported accuracy levels.

1 Introduction

Plant diseases represent a significant threat to global food security; early and accurate identification is essential for effective treatment and yield preservation. Manual diagnosis by trained agronomists is accurate but does not scale to the demands of modern agriculture. Automated classification systems based on computer vision offer a viable alternative.

This study evaluates six architectures on the PlantVillage dataset, a widely used benchmark for plant disease classification containing leaf images across 38 disease–plant categories. The objective is threefold: (1) to quantify the performance differences between CNN and transformer architectures under controlled conditions, (2) to analyse the trade-offs between classification accuracy, model complexity, and computational cost, (3) to assess the role of pre-training—supervised, self-supervised, or absent—in determining classification performance.

2 Related Work

The PlantVillage dataset was introduced by Hughes and Salathé [1] and has since become a standard benchmark for plant disease classification. Mohanty et al. [2] achieved 99.35% accuracy using transfer learning with GoogLeNet, establishing a strong baseline. Subsequent works have applied various architectures including VGG, ResNet, DenseNet, and Vision Transformers to this task. Too et al. [3] compared DenseNet, VGG-16, Inception V4, and ResNet with transfer learning and found DenseNet-121 to be the most effective, achieving 99.75% accuracy.

A recurring concern in the PlantVillage literature is data leakage due to the controlled acquisition conditions. Ferentinos [4] reported accuracies exceeding 99% but noted that performance drops significantly when models are evaluated on images from different sources. Most prior works employ transfer learning from ImageNet-pretrained weights. In contrast, this study trains the CNN models from scratch to evaluate raw architectural capacity without the confounding effect of pretrained feature representations.

The present study bridges these lines of work by comparing CNNs trained from scratch, a self-supervised ViT evaluated via linear probing (DINOv2), a supervised-pretrained ViT with selective fine-tuning, and a ViT-B/16 trained from random initialisation, all under controlled experimental conditions.

3 Dataset

3.1 Description

The PlantVillage dataset consists of approximately 54,300 colour images of plant leaves spanning 38 classes. Each class corresponds to a specific plant–disease combination (e.g., Tomato—Bacterial Spot, Apple—Cedar Apple Rust) or a healthy specimen, covering 14 plant species. Images are captured under controlled laboratory conditions with uniform back-

grounds and consistent lighting, and are stored as JPEG files at a resolution of 256×256 pixels. The dataset exhibits moderate class imbalance: the smallest class contains 152 samples while the largest contains 5,507, with a mean class size of approximately 1,429 images.

3.2 Data Splitting

Two splitting protocols were used across the experiments. For the CNN and DINOv2 models, the dataset was split into a training set (43,444 images, 80%) and a test set (10,861 images, 20%) using a random split with a fixed seed (42). No validation set was employed as the objective was comparative evaluation under default hyperparameters. For the ViT and CLIP-architecture models, a stratified random split allocated 80% to training (43,444 images), 10% to validation (5,430 images), and 10% to test (5,431 images), also with seed 42. While the different split ratios preclude exact numerical comparison between the two groups, the relative ranking and qualitative conclusions remain valid given the large dataset size and consistent class distributions across splits.

3.3 Preprocessing

Training images for the CNN and DINOv2 models are augmented with random resized crops to 224×224 pixels and random horizontal flips. Test images are resized to 256 pixels on the shorter edge and centre-cropped to 224×224. Both sets are normalised using the ImageNet channel-wise mean and standard deviation.

The ViT model additionally employs ColorJitter (brightness = 0.2, contrast = 0.2, saturation = 0.2) during training and uses a normalisation of mean $= [0.5, 0.5, 0.5]$, std $= [0.5, 0.5, 0.5]$. The CLIP-architecture model uses `open_clip`'s native preprocessing pipeline (resize, centre crop, normalisation) without additional augmentation.

3.4 Dataset Limitations

A known limitation of PlantVillage is the correlation between images within the same class. Images are acquired in sequential batches from the same leaf specimens, and consecutive images from the same source exhibit a mean pixel-wise absolute difference of approximately 30–35, compared to approximately 45 for images from different sources within the same class. Under a random train/test split, near-duplicate images of the same leaf may appear in both sets, leading to inflated performance estimates. A split stratified by leaf or acquisition source would yield more conservative and realistic generalisation estimates, with prior work employing such splits reporting accuracy reductions of 10–20 percentage points. The reported accuracies should be interpreted in this context.

4 Models

4.1 AlexNet

AlexNet [5] consists of 5 convolutional layers followed by 3 fully connected layers. It introduced the use of ReLU activations, dropout regularisation, and overlapping max pooling. The final fully connected layer is replaced with a linear layer mapping to 38 output classes. The model contains 57.2M parameters (218.1 MB).

4.2 VGG-16

VGG-16 [6] employs a uniform architecture of 13 convolutional layers using exclusively 3×3 filters, followed by 3 fully connected layers. This design demonstrated that depth with small receptive fields can substitute for larger filters while increasing representational capacity. With 134.4M parameters (512.8 MB), VGG-16 is the largest model in this study. The parameter count is dominated by the fully connected layers ($\sim$89%).

4.3 ResNet-50

ResNet-50 [7] introduced residual connections that enable gradient flow through very deep networks, mitigating the vanishing gradient problem. The architecture consists of 50 layers organised into 4 stages of bottleneck blocks, with batch normalisation after each convolutional layer. At 23.6M parameters (90.3 MB), ResNet-50 is the most parameter-efficient CNN model.

4.4 DINOv2 (ViT-B/14)

DINOv2 [8] is a self-supervised Vision Transformer trained using a combination of image-level and patch-level objectives on a curated dataset of 142 million images (LVD-142M). The ViT-B/14 variant processes 224×224 images as sequences of 256 non-overlapping 14×14 patches through 12 transformer encoder layers with 12 attention heads and an embedding dimension of 768. The pretrained backbone is used as a frozen feature extractor, and a single linear layer maps the 768-dimensional `[CLS]` token output to 38 classes. This linear probing protocol is the standard evaluation method for self-supervised models. The total model comprises 86.6M parameters (86.6M frozen backbone, 0.029M trainable head).

4.5 Vision Transformer (ViT-B/16, Fine-tuned)

The ViT model is based on `google/vit-base-patch16-224` [9], accessed via the Hugging Face Transformers library. The base configuration processes 224×224 images as 196 non-overlapping 16×16 patches through 12 transformer encoder layers with 12 attention heads, hidden dimension 768, and intermediate feed-forward

dimension 3072. A 38-class linear classification head replaces the original output layer.

To balance leveraging pre-trained representations with domain adaptation, a selective unfreezing strategy was adopted: all parameters were first frozen, then the classification head and the last 6 of the 12 encoder layers were unfrozen, yielding approximately 3.7M trainable parameters. This approach prevents catastrophic forgetting of low- and mid-level visual features learned during ImageNet pre-training while allowing high-level representations to specialise.

4.6 CLIP-Architecture ViT-B/16 (From Scratch)

The CLIP-based classifier uses the ViT-B/16 visual backbone from the `open_clip` library [10]. The architecture employs 12 transformer encoder layers, 12 attention heads, and a 512-dimensional visual embedding space. A custom module wraps the visual encoder with L2 normalisation and a linear classification head mapping to 38 class logits.

A critical aspect of this experiment is that the visual encoder was initialised with random weights (`pretrained=None`), meaning the training did not leverage CLIP's contrastive image–text pre-training. This experiment is therefore best understood as evaluating the ViT-B/16 architecture trained from scratch in a fully supervised setting, rather than as a transfer-learning experiment with CLIP representations. The full model occupies 571 MB on disk.

5 Experimental Setup

5.1 Training Configuration

The CNN models (AlexNet, VGG-16, ResNet-50) are trained from scratch for 30 epochs using SGD with momentum 0.9 and cosine annealing learning rate schedule. Model-specific hyperparameters follow the original publications: AlexNet uses learning rate 0.01, weight decay 5×10^{-4}, batch size 128; VGG-16 uses learning rate 0.01, weight decay 5×10^{-4}, batch size 64; ResNet-50 uses learning rate 0.1, weight decay 1×10^{-4}, batch size 128.

DINOv2 trains only the linear classification head for 30 epochs using SGD with momentum 0.9, learning rate 0.001, no weight decay, and batch size 128.

The ViT fine-tuned model is trained for 50 epochs using AdamW with learning rate 5×10^{-6}, linear warmup and decay scheduling, batch size 4, and FP16 mixed precision. Best checkpoint is selected based on validation performance.

The CLIP-architecture model is trained for 20 epochs using AdamW with constant learning rate 5×10^{-6}, batch size 4, and no learning rate scheduler.

All models use cross-entropy loss. Training is performed on GPU-equipped compute clusters (NVIDIA GTX 2080 and comparable hardware).

5.2 Evaluation Metrics

Models are evaluated on the held-out test set using: top-1 and top-5 accuracy, balanced accuracy, macro-averaged precision, recall, and F1 score, per-class ROC-AUC, predictive entropy, model size, and training/inference time.

6 Results

6.1 Classification Performance

Table 1 summarises the classification performance across all six models.

VGG-16 achieves the highest performance across all classification metrics. Its top-1 accuracy of 99.38% and balanced accuracy of 99.10% indicate near-perfect classification with consistent performance across classes. ResNet-50 ranks second (98.86%), followed by AlexNet (98.66%) and DINOv2 (98.37%). The fine-tuned ViT achieves 98.00%, while the randomly initialised CLIP-architecture model trails at 95.07%.

The gap between top-1 and balanced accuracy is most pronounced for AlexNet (0.83 pp) and DINOv2 (0.73 pp), suggesting greater sensitivity to class imbalance. VGG-16 exhibits the smallest gap (0.28 pp). All CNN, DINOv2 and ViT models achieve ROC-AUC values above 0.999, indicating near-optimal ranking ability.

6.2 Prediction Confidence

Predictive entropy provides a measure of model confidence. The fine-tuned ViT exhibits the lowest mean predictive entropy (0.0169 nats), followed closely by VGG-16 (0.0192 nats). AlexNet (0.0357) and ResNet-50 (0.0384) show comparable values. DINOv2 exhibits notably higher entropy (0.1499 nats)—approximately 4× higher than the CNN models—consistent with the linear probing constraint: because only a single linear layer is trained, the model has limited capacity to sharpen its probability distributions. The CLIP-architecture model shows the highest entropy (0.4869 nats), reflecting both the absence of pre-training and the overfitting observed during training. For reference, maximum entropy for a uniform distribution over 38 classes is $\ln(38) \approx 3.64$ nats; all models operate far below this bound.

The low entropy of the fine-tuned ViT is notable: despite having fewer trainable parameters than any CNN in this study, it produces the most confident predictions. A plausible explanation is that the frozen lower layers, pre-trained on ImageNet, provide well-calibrated intermediate representations, while the fine-tuned upper layers learn to produce sharply peaked class posteriors. This contrasts with DINOv2, where the frozen backbone is equally powerful but the single linear head lacks the capacity to concentrate probability mass as effectively.

Table 1: Classification performance across all models.

	AlexNet	VGG-16	ResNet-50	DINOv2	ViT (FT)	CLIP-arch
Top-1 Accuracy	98.66%	99.38%	98.86%	98.37%	98.00%	95.07%
Top-5 Accuracy	99.96%	99.99%	99.92%	99.97%	99.93%	99.26%
Balanced Accuracy	97.83%	99.10%	98.45%	97.64%	97%	92%
Precision (macro)	98.09%	99.00%	98.04%	97.86%	98%	95%
Recall (macro)	97.83%	99.10%	98.45%	97.64%	97%	92%
F1 Score (macro)	97.92%	99.05%	98.23%	97.74%	98%	93%
ROC-AUC (macro)	0.9998	0.9999	0.9997	0.9997	0.9999	0.9938
Pred. Entropy (nats)	0.0357	0.0192	0.0384	0.1499	0.0169	0.4869

Note: Two splitting protocols were used. For the CNN and DINOv2 models, which were evaluated under fixed hyperparameters without checkpoint selection, the dataset was split into training (80%) and test (20%) sets. For the ViT and CLIP-architecture models, which require a validation set for checkpoint selection and early stopping, a stratified 80/10/10 split was adopted.

6.3 Per-Class Analysis

Per-class analysis of the ViT and CLIP-architecture models reveals informative patterns. The fine-tuned ViT achieves F1 ≥ 0.92 on all 38 classes, with 7 classes at perfect F1 (1.00), including Corn Common Rust, Orange Huanglongbing, and Squash Powdery Mildew. Its weakest class is Potato Healthy (F1 = 0.89, $n = 19$).

The CLIP-architecture model shows greater variance across classes, with 6 classes below F1 = 0.92. Potato Late Blight is particularly problematic (F1 = 0.72, recall = 0.56), indicating that the model fails to identify more than 40% of its occurrences—likely due to visual similarity with Potato Early Blight and insufficient discriminative features learned under random initialisation.

Both models struggle most with minority classes, confirming that class imbalance is the primary per-class challenge. Among the CNN models, VGG-16's per-class AUC never falls below 0.997, while DINOv2's worst per-class AUC is 0.989 (Tomato—Early Blight). Figure 1 shows the per-class AUC distribution for VGG-16.

6.4 Model Complexity and Computational Cost

Table 2 presents the efficiency characteristics of each model.

ResNet-50 is the most parameter-efficient model, achieving the second-highest accuracy with only 23.6M parameters—2.4× fewer than AlexNet and 5.7× fewer than VGG-16. This efficiency is attributable to the bottleneck block design and global average pooling in place of large fully connected layers.

DINOv2 is the slowest model overall (54.3 s inference vs 10–15 s for CNNs), reflecting the higher computational cost of self-attention over convolutional operations at this image resolution. The CLIP-architecture model is notably larger on disk (571 MB) than the fine-tuned ViT (328 MB) despite lower accuracy, as the checkpoint includes text encoder components. AlexNet trains fastest (1,454 s) and has the lowest in-

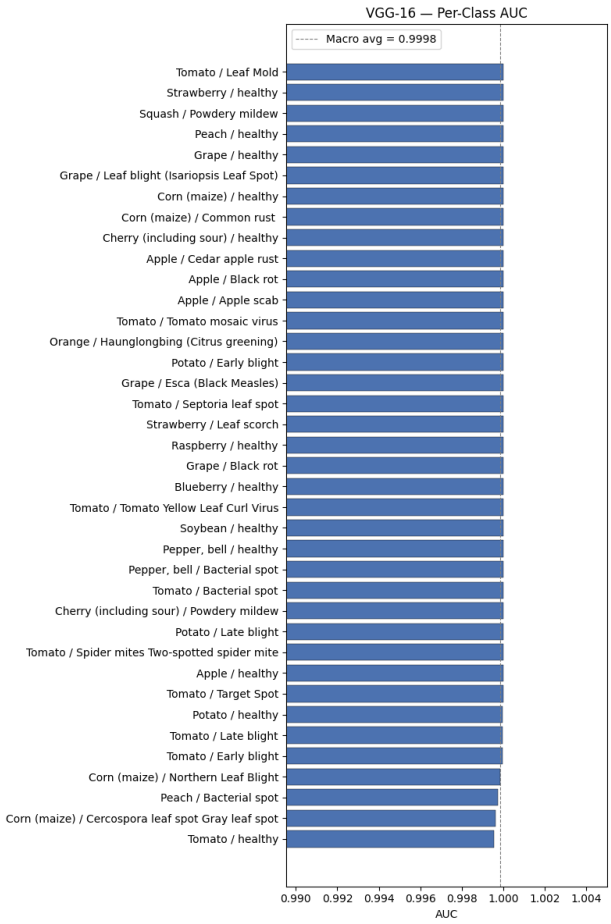

Figure 1: Per-class ROC-AUC for VGG-16. All 38 classes achieve AUC ≥ 0.997, with a macro average of 0.9998.

Table 2: Model complexity and computational cost.

	AlexNet	VGG-16	ResNet-50	DINOv2	ViT (FT)	CLIP-arch
Parameters (M)	57.2	134.4	23.6	86.6	∼86 (3.7)	∼87
Model Size (MB)	218.1	512.8	90.3	329.5*	328	571
Train Time (s)	1 454	5 102	2 459	6 390	–	–
Inference Time (s)	10.4	15.3	14.0	54.3	–	–

* DINOv2 model size reflects the frozen ViT-B/14 backbone (∼330 MB), which is publicly available and shared across tasks; only the linear classification head (0.11 MB, 29,222 parameters) is task-specific and stored separately.

ference latency (10.4 s).

6.5 Training Dynamics

Analysis of training curves reveals distinct convergence behaviours. Among the CNNs, AlexNet converges smoothly from loss 3.07 (epoch 1) to 0.11 (epoch 30). VGG-16 follows a similar trajectory, reaching 0.07 with slightly higher training accuracy (97.59%). ResNet-50 exhibits a higher initial loss (3.74) due to its higher learning rate (0.1), but converges to a comparable final loss (0.08) with training accuracy of 97.50%. The training curves for all four end-to-end trained models are shown in Figures 2 and 3.

DINOv2's linear head converges rapidly—training accuracy jumps from 81.45% (epoch 1) to 92.77% (epoch 2)—reflecting the high quality of the pretrained features. However, it plateaus at 96.87% training accuracy with a final loss of 0.1156, suggesting that some class distinctions are not linearly separable in DINOv2's feature space.

The fine-tuned ViT shows consistent improvement over 50 epochs, rising from 65.27% validation accuracy (epoch 1) to 97.51% (epoch 50), with the learning rate decaying smoothly via linear scheduling (Figure 4).

The CLIP-architecture model exhibits rapid early convergence (80.35% at epoch 1, 97.16% validation accuracy at epoch 15) but shows signs of overfitting thereafter, declining to 95.32% validation accuracy at epoch 20 (Figure 5). The gap between final training accuracy (99.20%) and validation accuracy (95.32%) is consistent with the absence of a learning rate scheduler and data augmentation.

7 Discussion

7.1 Architectural Trade-offs

The results reveal a clear trade-off between model complexity and performance. VGG-16 achieves the best accuracy but at the highest computational cost among the CNNs. ResNet-50 offers the best parameter efficiency, achieving comparable performance with 5.7× fewer parameters. AlexNet, despite being the simplest and fastest model, remains competitive—trailing VGG-16 by only 0.72 pp.

The relatively small performance gap among the top four models (98.37–99.38%) is likely attributable to the nature of the dataset. The controlled acquisition conditions, uniform backgrounds, and limited intra-class variability make PlantVillage a relatively easy classification task. Under these conditions, even the simplest architecture captures sufficient discriminative features. The advantages of deeper architectures are more likely to manifest on datasets with greater visual complexity, background clutter, or domain shift.

7.2 The Role of Pre-training

A central finding of this study is that training paradigm—not architecture—is the dominant factor determining classification performance.

DINOv2 (ViT-B/14), despite using a powerful self-supervised backbone pretrained on 142 million images, achieves the lowest accuracy among the top four models (98.37%). This result reflects the linear probing constraint: the single-layer classifier cannot capture nonlinear feature combinations, and the higher predictive entropy (0.1499 vs 0.02–0.04 for CNNs) confirms less confident predictions. However, 98.37% accuracy with only 29,222 trainable parameters is remarkable given that the backbone has never seen a plant disease image during pretraining.

The comparison between the fine-tuned ViT (98.00%) and the randomly initialised CLIP-architecture ViT-B/16 (95.07%) provides direct evidence that pretraining confers a measurable advantage. Both share the ViT-B/16 backbone topology, but the 2.93 pp accuracy gap and 0.05 macro-F1 gap confirm that supervised ImageNet pre-training translates into superior performance and robustness, particularly on minority and visually ambiguous classes.

The CLIP-architecture model's results also demonstrate, however, that the ViT-B/16 architecture has sufficient capacity to reach 95.07% accuracy from random initialisation on this dataset, underscoring the architecture's intrinsic representational power.

7.3 CNN vs Transformer Performance

A perhaps surprising result is that all three CNNs trained from scratch outperform DINOv2 via linear probing and the fine-tuned ViT. This outcome reflects two factors: (1) end-to-end training enables all network layers to co-adapt for the specific task, which is especially advantageous when the dataset has limited visual complexity and the discriminative features are

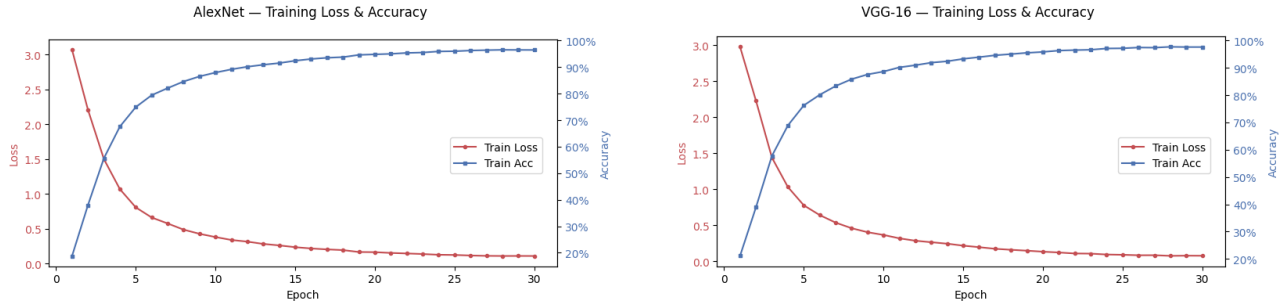

Figure 2: Training loss and accuracy for AlexNet (left) and VGG-16 (right). Both models exhibit smooth convergence over 30 epochs under cosine annealing.

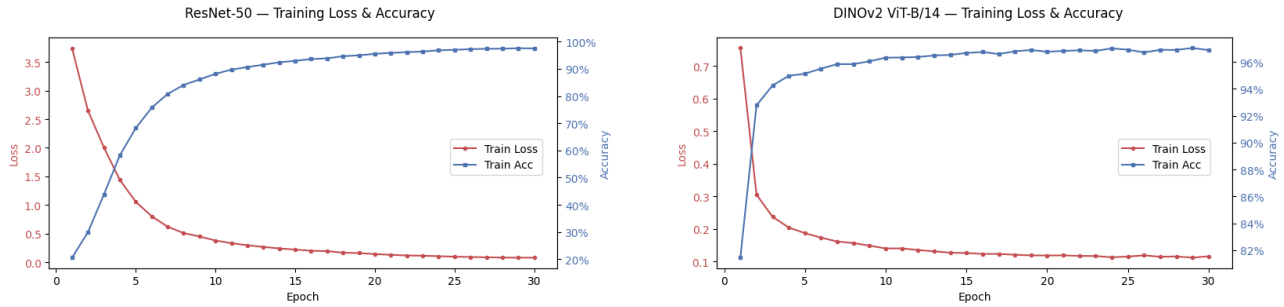

Figure 3: Training loss and accuracy for ResNet-50 (left) and DINOv2 linear head (right). ResNet-50 starts with higher loss due to its larger initial learning rate. DINOv2's linear head converges rapidly but plateaus, suggesting limited linear separability of some class boundaries.

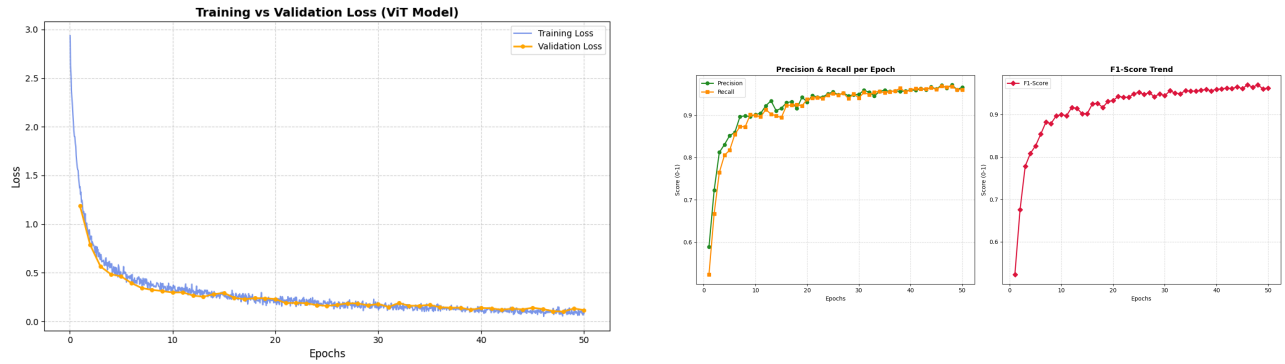

Figure 4: Fine-tuned ViT training dynamics. Left: training and validation loss over 50 epochs, showing smooth convergence without overfitting. Right: macro precision, recall, and F1 score over epochs, all converging above 0.95.

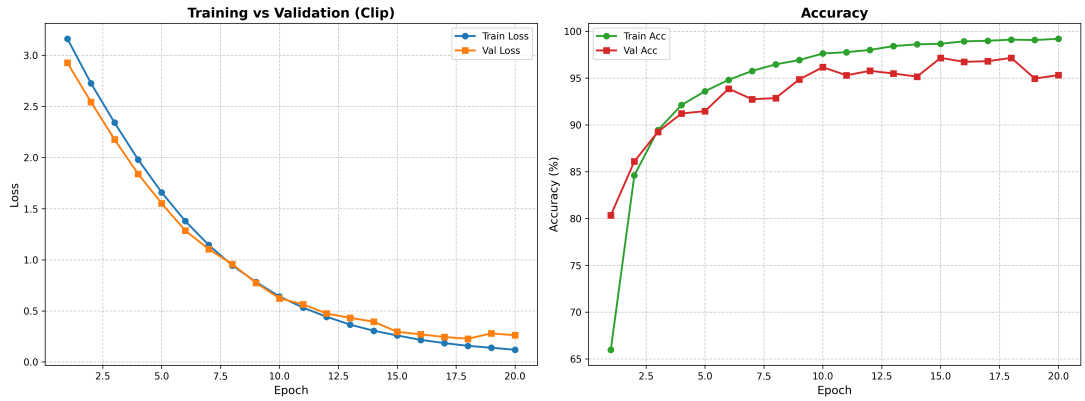

Figure 5: CLIP-architecture ViT-B/16 training dynamics. The divergence between training and validation accuracy after epoch 15 indicates overfitting, consistent with the absence of a learning rate scheduler.

learnable from the training data alone; and (2) the linear probing and selective fine-tuning protocols intentionally constrain the transformer models' adaptability. Fine-tuning the DINOv2 backbone—rather than freezing it—would likely close or reverse the performance gap but falls outside the linear probing protocol used here.

7.4 On the Observed Accuracy Levels

The top-1 accuracies reported here (95.07–99.38%) are consistent with published results on PlantVillage under random splitting. However, these numbers should not be interpreted as indicative of field-deployable performance. The dataset's controlled conditions differ substantially from real agricultural settings, where images exhibit variable lighting, complex backgrounds, multiple diseases per leaf, and different camera characteristics. The presence of near-duplicate images from sequential acquisitions means the effective diversity of the test set is lower than the sample count suggests. A leaf-level split would provide a more realistic estimate of generalisation, with prior work reporting accuracy reductions of 10–20 pp under such protocols.

7.5 Training Dynamics and Overfitting

The CLIP-architecture model is the only model exhibiting clear overfitting, with validation accuracy peaking at epoch 15 and declining thereafter. This is consistent with the absence of a learning rate scheduler and limited data augmentation. All other models show stable convergence within their respective training horizons, with cosine annealing (CNNs, DINOv2) and linear decay (ViT) contributing to smooth final-epoch behaviour.

8 Conclusion

This study provides a controlled comparison of six architectures on the PlantVillage plant disease classification task. The key findings are as follows:

1. VGG-16 achieves the highest top-1 accuracy (99.38%) among all models, while ResNet-50 offers the most favourable accuracy-to-efficiency ratio (98.86% with 23.6M parameters). AlexNet remains competitive despite its simpler design.

2. CNNs trained from scratch outperform both the linearly probed DINOv2 and the fine-tuned ViT on this benchmark. This reflects the advantage of end-to-end training on a task with limited visual complexity, rather than an inherent architectural superiority.

3. Pre-training provides a consistent advantage for transformers. The fine-tuned ViT (98.00%) outperforms the randomly initialised ViT-B/16 (95.07%) by 2.93 pp, confirming that supervised pre-training translates into better performance and robustness on downstream tasks.

4. DINOv2's self-supervised features transfer well to this domain, achieving 98.37% accuracy with only a linear head, but a single linear layer does not fully exploit the backbone's capacity.

5. Class imbalance remains the primary per-class challenge across all models, with minority classes (especially Potato Healthy) consistently underperforming.

6. Model size does not predict performance. The CLIP-architecture model (571 MB) is the largest yet achieves the lowest accuracy, while ResNet-50 (90.3 MB) outperforms models several times its size.

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